# Jinwoo Jung

M.S. student, Hanyang University | ASML Lab, advised by Prof. Joon-Hyuk Chang

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## Research Interests

Diffusion and flow-based generative models; controllable generation and editing; training-free inference efficiency.

## Education

**Hanyang University**, Seoul, Republic of Korea — Mar 2025 – Feb 2027 (expected)
M.S. in Artificial Intelligence Semiconductor Engineering
- GPA 4.44/4.50.

**Konkuk University**, Seoul, Republic of Korea — Mar 2019 – Feb 2025
B.S. in Computer Science and Engineering
- GPA 3.88/4.50 (major 3.97/4.50).

**Zurich University of Applied Sciences (ZHAW)**, Winterthur, Switzerland — Sep 2022 – Jan 2023
Exchange student, School of Engineering

## Publications

* denotes equal contribution; † denotes corresponding author.

**SARA: Step-Adaptive Rank Adjustment for Diffusion Inference Acceleration** — NeurIPS 2026
Dong Hoon Lim*, **Jinwoo Jung***, Won-Gook Choi, Joon-Hyuk Chang† [paper] [code]

**Edit the Moment, Keep the Rest: Time-Localized Audio Editing via Instruction** — INTERSPEECH 2026
**Jinwoo Jung**, Gihun Son, Won-Gook Choi, Joon-Hyuk Chang† [paper] [demo]

**Branch-wise Complementary Attention for Acoustic Scene Classification** — INTERSPEECH 2026
Seung-Gyu Han*, **Jinwoo Jung***, Pil Moo Byun, Won-Gook Choi, Joon-Hyuk Chang† [paper]

**Machine Learning Algorithm-Based Prediction Models for Adolescent Suicide Attempt** — IEEE ICCE 2025
**Jinwoo Jung**, Seulmi Kang, Myeonghwan Go, Sieon Lee, Seokheon Cho† [paper]

## Manuscripts

**ENCORE: Training-Free Few-Step Diffusion Sampling with Terminal Moment Matching** — under review, 2026
**Jinwoo Jung**, Dong Hoon Lim, Joon-Hyuk Chang†, Defang Chen

**TRACE: Trajectory-Conditioned Editing of Sound Sources in First-Order Ambisonics** — under review, 2026
**Jinwoo Jung**, Dong Hoon Lim, Joon-Hyuk Chang†

## Presentations

**NeurIPS 2026**, Atlanta, USA — SARA (poster, presenting author) — Dec 2026 (upcoming)
**INTERSPEECH 2026**, Sydney, Australia — EMKR (poster, presenting author) and BCA (poster) — Sep 2026

## Patents

All applications are filed in the Republic of Korea.

**System and Method for Accelerating Diffusion Model Inference Using Step-Adaptive Low-Rank Adjustment** — filed Oct 2026

**Branch-wise Complementary Attention for Acoustic Scene Classification** — filed Oct 2026

**Interval Conditioning and Source-Event Masking for Instruction-based Time-Localized Audio Editing** — filing in progress

## Research Experience

**Hanyang University**, Seoul, Republic of Korea — Mar 2025 – present
ASML Lab (Acoustic, Speech Signal Processing and Machine Learning)
- AI Model Compression and Acceleration, with Samsung Electronics (Jan 2026 – Sep 2026).
- Sound-Tracking AI Agent via Connected Devices, with LG Electronics (Jun 2025 – Sep 2025).

**Qualcomm Institute**, UC San Diego, La Jolla, United States — Jan 2024 – Feb 2024
Artificial Intelligence (AI) Development Project
- Team lead; our team placed first.
- The team project continued after the program and became a conference paper (IEEE ICCE 2025).

## Awards

1st Place, IC-PBL+ Research Showcase, Hanyang University — Jun 2026
Outstanding Achievement Award, Qualcomm Institute AI Development Project, UC San Diego — Feb 2024

## Skills and Languages

**Programming** Python, PyTorch, librosa, torchaudio | **Languages** Korean (native), English (TOEFL iBT 100)